CareBridge NB — Dataset Catalog

Nov 11, 2026 · @Jason

Overview

CareBridge NB uses 18 sources, all free to download with no access request: 10 datasets the app queries, 5 rule sets and links for eligibility, and 3 training sets for the two models the project trains (an urgency classifier and a service matcher). The lookup data is never used for training: facts like phone numbers and income limits stay in tables and rules, and the models learn only from free text.

Selection criteria: each source supports at least one app feature, can be filtered to New Brunswick / Middlesex County (or applies to every NJ resident), and is under 3 years old unless its age doesn't matter.

Source	Type	Main app feature	Refresh
CMS Hospital General Information	Lookup data	Hospital match and ranking	Quarterly
CMS HCAHPS Patient Survey	Lookup data	Hospital match (trust signal)	Quarterly
CMS Timely and Effective Care	Lookup data	Emergency routing (ER times)	Quarterly
Hospital price transparency files	Lookup data	Insurance-network match, self-pay cost	Yearly
HRSA Health Center Sites	Lookup data	Low-cost clinic finder	Weekly
SAMHSA FindTreatment locator	Lookup data	Mental health and substance-use finder	Yearly
NPPES NPI Registry	Lookup data	Provider finder by specialty (dental, pediatric, OB-GYN)	Monthly
CMS State-based Exchange plan file	Lookup data	Marketplace plan comparison	Yearly
NJ DOBI carrier enrollment report	Lookup data	Local insurer ranking, Medicaid plan options	Quarterly

Source	Type	Main app feature	Refresh
Census ACS (S1701, S2701) and SAHIE	Lookup data (optional)	Community dashboard	Yearly
HHS Poverty Guidelines	Rules	Eligibility checker	Yearly
NJ FamilyCare income limits	Rules	Eligibility checker	Yearly
Cover All Kids	Rules	Eligibility checker (children)	When law changes
NJ Charity Care rules	Rules	Eligibility checker, cost estimate	Yearly
Get Covered NJ	Link-out	Plan enrollment and subsidies	Yearly
NHAMCS Emergency Department file, 2022	Training data	Urgency classifier (real triage labels)	Fixed
Symptom-to-diagnosis text set	Training data	Patient-language examples for both models	Fixed
AI-generated labeled messages	Training and test data	Both models	You maintain it

Hospital data

Four sources make up the `hospitals` table. They join on `Facility ID` (the CMS certification number), filtered to Middlesex County.

Dataset	What it contains	Example columns	How the app uses it
CMS Hospital General Information (`xubh-q36u`)	One row per hospital: identity, location, type, ownership, overall quality	`Facility ID`, `Facility Name`, `Address`, `City/Town`, `ZIP Code`, `County/Parish`, `Telephone Number`, `Hospital Type`, `Hospital Ownership`, `Emergency Services`, `Hospital overall rating`	**Hospital match and ranking.** It's the base list. `Emergency Services` is the first filter for urgent needs, and the star rating is a ranking signal.
CMS HCAHPS Patient Survey (`dgck-syfz`)	Patient-reported experience: one row per hospital per survey question	`Facility ID`, `HCAHPS Measure ID`, `HCAHPS Question`, `HCAHPS Answer Percent`, `Patient Survey Star Rating`, `Number of Completed Surveys`	**Trust signal.** Pivot to one row per hospital, then show lines like "Patients rate communication 4/5."
CMS Timely and Effective Care (`yv7e-xc69`)	Process measures, including emergency department timing; one row per hospital per measure. CMS lists it as last modified 2026-07-22.	`Facility ID`, `Condition`, `Measure ID`, `Measure Name`, `Score`, `Sample`, `Start Date`, `End Date`. Useful measures: `OP_18b` (median ER time), `OP_22` (left before being seen), `EDV` (ER volume)	**Emergency routing.** Pick a nearby ER and set expectations: "The median ER visit here is about X minutes."

Dataset	What it contains	Example columns	How the app uses it
Hospital price transparency files	Each hospital's own machine-readable file of standard charges for all items and services, published on its website in the CMS template format	`hospital_name`, `description`, `code`, `setting`, `gross_charge`, `discounted_cash`, `payer_name`, `plan_name`, `standard_charge_methodology`, `negotiated_dollar`, `minimum`, `maximum`	**Insurance-network match and self-pay cost.** `payer_name` and `plan_name` show which insurers a hospital has contracts with. `discounted_cash` gives uninsured users a price.

Filtering

- **General Information:** `State` = NJ and `County/Parish` = Middlesex, then keep hospitals within about 15 miles of New Brunswick. That's the list of `Facility ID`s every other hospital table is filtered to.
- **HCAHPS:** keep those `Facility ID`s and only the measures you show (overall star, would recommend, nurse and doctor communication), then pivot to one row per hospital.
- **Timely and Effective Care:** keep those `Facility ID`s and only the emergency measures (`OP_18b`, `OP_22`, `EDV`).
- **Price transparency:** download the files of the local hospitals only, keep a short list of common services (ER visit levels, labor and delivery, basic imaging), then take the unique payer and plan pairs.

Notes

- The price transparency files are huge. Keep only the rows for your 2–4 local hospitals and a short list of common services (ER visit levels, labor and delivery, imaging). Then take the unique `payer_name` and `plan_name` pairs per hospital.
- None of these datasets say whether a hospital has a pediatric ER or a children's hospital. Add `has_pediatric_er`, `has_childrens_hospital` and `pediatric_specialties` by hand from hospital websites.

- Check the general-information and HCAHPS column names against your download. They come from the CMS data dictionaries, not this research.

Clinics and low-cost care

For uninsured users with non-emergency needs, a health center is usually a better answer than a hospital. This section becomes the `clinics` table.

Dataset	What it contains	Example columns	How the app uses it
HRSA Health Center Service Delivery and Look-Alike Sites	Over 16,200 sites where HRSA-funded health centers and look-alikes provide care. Look-alikes serve everyone regardless of ability to pay. CSV or XLSX download.	`Site Name`, `Site Address`, `Site City`, `Site Postal Code`, `Site Telephone Number`, `Health Center Type`, `Health Center Location Type` (Permanent / Mobile Van), `Health Center Location Setting` (e.g. School), `Operating Hours per Week`, county, latitude/longitude	**Low-cost clinic finder.** Non-urgent and uninsured users go here first. Sort by distance and highlight sliding-fee care.

Dataset	What it contains	Example columns	How the app uses it
SAMHSA FindTreatment.gov locator	Mental health and substance-use treatment facilities. It's a public JSON API (`/locator/exportsAsJson/v2`) that searches around a lat/long. Updated annually from a national facility survey.	Facility name, street, city, ZIP, phone, latitude, longitude, plus service codes for care type, payment accepted (Medicaid, sliding fee, free) and languages	**Mental health and substance-use finder.** Filter to sliding-fee or Medicaid-accepting facilities near New Brunswick. Always show 988 alongside.
Local free and student-run clinics	No federal dataset covers these	Build your own: `name`, `address`, `hours`, `services`, `languages`, `walk_in`, `documents_needed`	**Low-cost clinic finder.** Covers care HRSA misses. Enter by hand and re-check every semester.

Filtering

- **HRSA:** state = NJ, county = Middlesex, then keep active sites within about 15 miles of New Brunswick by lat/long.
- **SAMHSA:** query the API with New Brunswick's coordinates and a radius (about 15 miles), then keep facilities whose payment codes include Medicaid, sliding fee or free care.

Notes

- Check HRSA column names against the metadata file that comes with the download. The ones above are from the published layout, not checked against this year's file.
- Dental care: many HRSA health centers offer it, so add a `dental` flag when you verify each local site.

Providers and community services

Hospitals and health centers don't cover dentists, private pediatricians, pharmacies or non-medical help. These sources fill that gap and become the `providers` and `services` tables.

Dataset	What it contains	Example columns	How the app uses it
NPPES NPI Registry	Every US healthcare provider's identifier, specialty (taxonomy) and practice address. Free CMS API with no key, searchable by specialty, city, state and ZIP, up to 1,200 results per query. Also a monthly bulk file.	`number` (NPI), `enumeration_type`, `basic.organization_name`, `taxonomies[].desc`, `addresses[].address_1`, `addresses[].city`, `addresses[].postal_code`, `addresses[].telephone_number`	**Provider finder.** Dentists, pediatricians, OB-GYNs and pharmacies near New Brunswick. It doesn't say who takes Medicaid or uninsured patients, so add `accepts_medicaid` and `sliding_fee` flags by hand for the ones you list.
CMS Doctors and Clinicians (`mj5m-pzi6`)	Medicare-enrolled clinicians with specialty, practice address and a telehealth flag	`NPI`, `pri_spec`, `sec_spec_all`, `Telehlth`, `org_nm`, address and city columns	**Optional.** Joins to NPPES on `NPI` to add a telehealth flag. Medicare-only, so it's not a full provider list.

Dataset	What it contains	Example columns	How the app uses it
Local services list (hand-built)	Non-medical help in New Brunswick: food pantries, housing help, transport, prescription assistance. Built from the public websites of local organizations and county agencies.	`service_id`, `name`, `description`, `category`, `eligibility`, `address`, `hours`, `languages`, `source_url`	**Non-medical services** and the **service matcher's catalog.** The matcher embeds each `description`. Aim for 50–150 entries.

Notes

- **Filtering:** query NPPES by New Brunswick-area ZIPs (08901, 08902, 08903, 08904 and nearby towns) with one query per specialty, then keep active NPIs only. Filter Doctors and Clinicians to NJ and those same ZIPs.
- Check the NPPES field names and the `Telehlth` column name against a live response and the data dictionary.
- Re-check the hand-built services list every semester and keep `source_url` so each entry can be verified.

Insurance and coverage

NJ runs its own exchange (Get Covered NJ). The federal HealthCare.gov plan files exclude state-based exchanges like NJ's, so plan data comes from the CMS state-based exchange file instead.

Dataset	What it contains	Example columns	How the app uses it
CMS State-based Exchange plan file (SBE QHP PUF)	Plan and issuer details for health and dental plans sold through state-based exchanges, NJ included	Plan ID, issuer name, plan marketing name, metal level, plan type (HMO/PPO/EPO), rating area or county, premium by age, deductible, out-of-pocket max	**Marketplace plan comparison.** For users who earn too much for NJ FamilyCare, show the cheapest Bronze, Silver and Gold plans in Middlesex. Get the current plan year.
NJ DOBI carrier enrollment report, Q2 2026	Quarterly enrollment counts by county and insurer, split into commercial and Medicaid. Insurers include Horizon, Oscar, Oxford, UnitedHealthcare, CIGNA, AmeriHealth and WellCare.	One row per insurer × type (`Comm Net`, `Comm Non Net`, `Medicaid`, `Total`) and one column per county (`Middlesex` ...). For Middlesex: about 235,700 enrolled in total, 96,103 commercial and 139,312 Medicaid.	**Local insurer ranking.** Decides which insurers to look up first in the price transparency files. **Medicaid plan options:** its Medicaid rows show which NJ FamilyCare plans operate locally.
Get Covered NJ	The official enrollment site, with subsidy information and open enrollment dates	No download. Link to it.	**Plan enrollment.** After plan comparison, send users here to actually sign up.

Filtering

- **SBE plan file:** state = NJ, the current plan year, the individual market (not small business), and the rating area or service area that covers Middlesex County.
- **DOBI report:** keep the `Middlesex` column only, and skip insurers whose Middlesex count is zero.

Notes

- The DOBI report is a PDF. Parse it with `pdfplumber`, or type in the Middlesex column by hand since that's all you need. Its counts also include out-of-state residents who work in NJ.
- Plan file column names change slightly every year. Read the data dictionary that comes with the year you download.

Eligibility rules

These are published rules, not datasets. Code them into Python rules in an `eligibility_rules` config and update them every January. Everything is measured as a percentage of the HHS poverty guidelines (the federal poverty level).

Rule source	What it sets	Values to code (2026)	How the app uses it
HHS Poverty Guidelines	Base income per household size. In effect since Jan 13, 2026. ASPE also offers a Poverty Guidelines API.	`fpl_base = 15960` (1 person), `fpl_per_extra = 5680`	**Eligibility checker.** Turns household size and income into `fpl_percent`, which every other rule uses.
NJ FamilyCare (Medicaid / CHIP)	Income limits by group	Children under 19: ⩽ 355% FPL ($9,763/month for a family of 4). Adults 19–64: ⩽ 138% FPL ($1,836/month for 1). Pregnant: ⩽ 205% FPL	**Eligibility checker.** The first program to test, since it's full coverage.
Cover All Kids	Children under 19 qualify regardless of immigration status if they meet the income rule	`child_immigration_check = false`	**Eligibility checker (children).** Never ask about a child's immigration status. Only apply the income test.

Rule source	What it sets	Values to code (2026)	How the app uses it
NJ Charity Care (Hospital Care Payment Assistance)	Free or reduced-cost care at NJ acute care hospitals	⩽ 200% FPL: free. Patient pays 20% / 40% / 60% / 80% of charges in the >200–225 / 225–250 / 250–275 / 275–300% bands. Over 300%: 100%. Also requires being ineligible for other coverage and meeting an asset limit.	**Eligibility checker and cost estimate.** "At your income you'd pay about 40% of the hospital bill." Also covers people with insurance who still owe a balance.

Notes

- Medicaid rules are changing starting Fall 2026 (work requirements, more frequent renewals). Check the NJ FamilyCare site before you hard-code the adult rules.
- The adult and pregnancy limits above come from a secondary site that cites the official NJ FamilyCare income chart. Confirm them against the chart.
- Asset limits: an RWJBarnabas notice lists $7,500 per individual and $15,000 per family. That's a 2021 document, so confirm it with the current fact sheet.

Community context

Optional. These feed a community dashboard and the app's "why this matters" page. They never drive matching: eligibility always comes from what each user reports about themselves.

Dataset	What it contains	Example columns	How the app uses it
Census ACS S1701, Poverty Status	Share of residents below poverty by age, sex and work status, down to census tract	Tract `GEO_ID`, `NAME`, `S1701_C01_001E` (population counted), `S1701_C03_001E` (% below poverty)	**Community dashboard.** A map of New Brunswick tracts by poverty rate.

Dataset	What it contains	Example columns	How the app uses it
Census ACS S2701, Health Insurance Coverage	Insured and uninsured counts and rates by age, income and citizenship, down to census tract	Tract `GEO_ID`, `NAME`, `S2701_C05_001E` (% uninsured), with age and income breakdowns	**Community dashboard.** Shows where uninsured residents are concentrated. Can also guide outreach, like which areas to put flyers in.
Census SAHIE	Model-based uninsured estimates for every county, by age, sex and income-to-poverty ratio. The 2024 estimates came out Aug 6, 2026.	`statefips`, `countyfips`, `agecat`, `sexcat`, `iprcat`, `NUI` (uninsured count), `PCTUI` (% uninsured)	**Community dashboard.** Middlesex uninsured rate for the low-income groups the app targets, compared with the NJ average.

Filtering

- **ACS S1701 and S2701:** request NJ, Middlesex County, at census tract level, then keep the tracts inside New Brunswick.
- **SAHIE:** NJ and Middlesex County (`countyfips` 023), latest year, all ages under 65, and the income groups at or below 400% of the poverty level.

Notes

- Pull ACS through the free Census API (5-year estimates for tract-level data).
- Check the variable IDs above against the table's variable list before you use them.

Trained models

The project trains two systems, both on the free-text part of intake. That's the one thing the questionnaire can't capture. Insurance, income and household size come from the questionnaire and go to the rules engine, so no model predicts them. Urgency is split into two stages so the urgency labels come from real triage nurses (NHAMCS), and AI-generated text is used only to teach the model how people describe symptoms.

Model	Task	Training data	Baseline → final model	Metric that matters
Urgency, stage 1: reason classifier	Message → one of about 25 reason-for-visit groups (chest pain, fever, abdominal pain, injury, mental health …)	AI-generated patient-style messages for each group, plus the symptom text set relabeled to the same groups	TF-IDF + logistic regression → fine-tuned DistilBERT	Macro-F1
Urgency, stage 2: urgency model	Reason group + age (+ pain score from the questionnaire) → `emergency` / `urgent` / `routine`	NHAMCS visits with real triage levels (immediate/emergent → emergency, urgent → urgent, semi-urgent/non-urgent → routine)	Logistic regression → gradient boosting	Emergency recall, calibration
Service matcher	Message → top 5 services in your catalog. The care type (dental, pediatric, mental health) comes from the matched service.	AI-generated pairs of a user query and the correct service, built from your `clinics`, `providers` and `services` tables	Off-the-shelf sentence-transformer (`all-MiniLM-L6-v2`) → the same model fine-tuned with contrastive loss	Recall@5 and MRR
Ranker (later)	Reorders results by what users found helpful	Click and helpful/not-helpful logs from real use	Logistic regression → LightGBM	NDCG. Needs real users first.

Build order

1. Download the NHAMCS 2022 ED file. Keep visits with a recorded triage level, group the reason-for-visit codes into about 25 groups, and map triage levels to the 3 classes.
2. Train the urgency model on reason group + age. Check emergency recall and calibration, then compare against gradient boosting.

3. Generate patient-style messages with an LLM: about 40 per reason group, across ages and writing styles (short, rambling, misspelled, Spanish-English mix).
4. Generate the 300-message test set separately, with different prompts, so it doesn't share phrasing with the training messages.
5. Train the reason classifier: baseline first, then fine-tune DistilBERT.
6. Build the service catalog, generate 10–20 queries per service, fine-tune the matcher, and compare recall@5 with the off-the-shelf model.
7. Serve everything through FastAPI as `/triage` and `/match`.

Safety rules

- A keyword check ("chest pain", "can't breathe", "overdose") runs before the models and always routes to 911 and the nearest ER.
- The models can raise urgency but never lower it, and the app never tells anyone they don't need care.
- Set the decision threshold for high emergency recall, and accept more false alarms.
- NHAMCS triage levels were assigned with vital signs and an exam. The urgency model only sees the reason, age and pain score, so report that as a known limit.

Training data

These are the only sources the models learn from. None is specific to New Brunswick, and none needs to be: how urgent a symptom is doesn't depend on the city.

Dataset	What it contains	Example columns	Which model it trains
NHAMCS Emergency Department public use file, 2022	A national sample of emergency visits from CDC, free to download with no approval. Each visit has coded reasons for visit and a triage level from immediate to non-urgent. 2022 was the last survey year.	`RFV1` – `RFV3` (reason-for-visit codes), `IMMEDR` (triage level), `AGE`, `PATWT` (visit weight)	**Urgency model (stage 2).** The only source of real triage labels in the project.

Dataset	What it contains	Example columns	Which model it trains
Symptom-to-diagnosis text set (Hugging Face)	1,065 symptom descriptions written the way a patient talks, labeled with 22 diagnoses. Apache 2.0 license.	`input_text`, `output_text`	**Reason classifier (stage 1)**, as extra patient-style text.
AI-generated labeled messages	All training messages for the reason classifier and the service matcher, plus the 300-message test set. **These are AI-generated**, not written by real users.	`text`, `reason_group`, `urgency`, `service_id`, `prompt_id`, `split` (train / test)	**Reason classifier and service matcher.** The test set is what every result gets reported on.

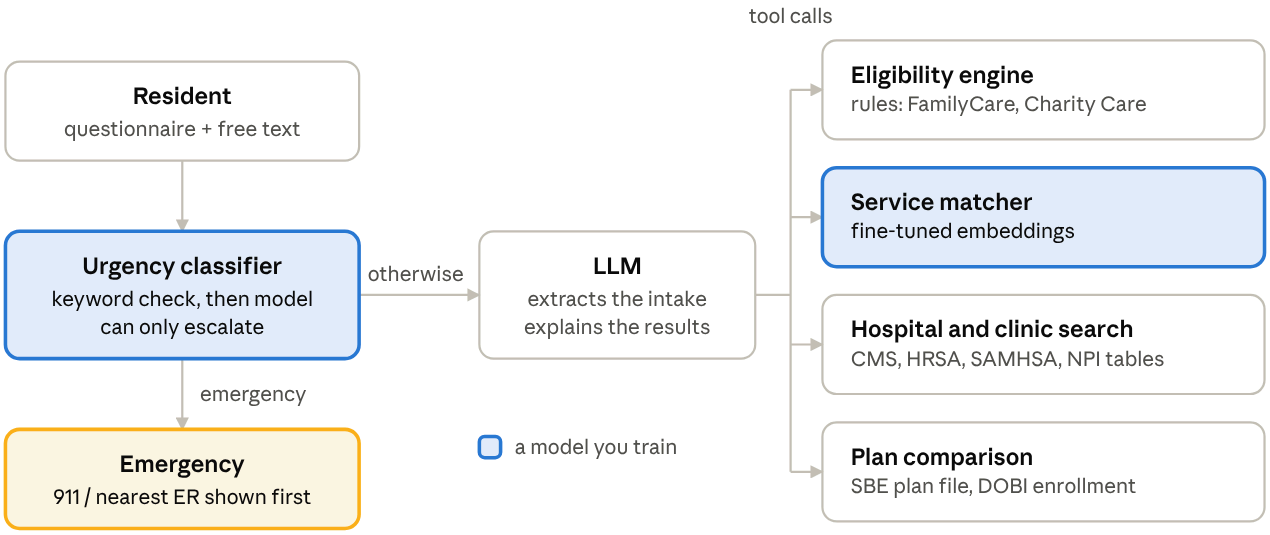
Notes

- **Filtering, NHAMCS:** drop visits with no triage level (blank, unknown, or "no triage" codes) and administrative visits (follow-ups, prescription refills). Combining the 2019–2022 files gives a bigger sample, if a single year is too small for rare reason groups.
- **Filtering, symptom text set:** drop diagnoses that don't map to any reason group, then relabel each row from its diagnosis to a reason group.
- **AI-generated data:** spot-check a random sample of labels by hand (about 50 messages) and log which prompt made each one (`prompt_id`). Because the test set is also AI-generated, scores will likely overstate how well the models do on real users. Say so whenever you report results, and swap in real messages once the app has users.
- NHAMCS is older than 3 years. That's fine here: symptoms and their urgency don't go stale the way prices and income limits do.
- Check the NHAMCS variable names against the 2022 file documentation.

How it fits together

Four parts share the work. The urgency classifier reads the free text first. The rules engine decides eligibility from the questionnaire. The service matcher and the search tools find where to go. The LLM runs the conversation and explains the results, and every fact it states comes from a tool.

Two trained models sit beside the rules engine and the LLM



CareBridge NB architecture · LLM, emergency check, 4 tools

Routing order: emergency → 911 and the nearest ER first; non-urgent and uninsured → clinics first; then NJ FamilyCare and Charity Care eligibility; insured → hospitals that work with their insurer. The optional community dashboard reads the Census tables directly and isn't part of this flow.

Table	Built from	Joined on
`hospitals`	CMS General Info + HCAHPS (pivoted) + Timely and Effective Care (ER rows) + manual pediatric columns	`Facility ID`
`hospital_payers`	Price transparency files (unique payer and plan pairs per hospital, plus cash prices for common services)	`Facility ID` / hospital name

Table	Built from	Joined on
`clinics`	HRSA sites + SAMHSA locator + manual free-clinic list	Lat/long for distance
`providers`	NPPES NPI Registry (+ optional telehealth flag from Doctors and Clinicians) + manual Medicaid and sliding-fee flags	`NPI`
`services`	Hand-built local services list. One row per thing a resident can be sent to, with a text description the matcher embeds	`service_id`, linked to `clinics` / `providers` / `hospitals`
`plans`	CMS SBE plan file (NJ, Middlesex)	Issuer name ↔ `insurers_local`
`insurers_local`	DOBI enrollment report (Middlesex column)	Issuer name
`eligibility_rules`	Poverty guidelines, NJ FamilyCare, Cover All Kids, Charity Care	Code, not a table
`training_examples`	NHAMCS, symptom text set, AI-generated messages. Kept apart from the app database.	`service_id` for matcher pairs
`area_stats` (optional)	ACS S1701, S2701, SAHIE	Census tract / county FIPS